# E-Signature Process Overview

---

## Current Flow (Document Preparation)

1. Documents Upload
2. Recipient Add
3. Signature Fields
4. Security
5. Settings
6. Review

---

## Updated Flow (Document Preparation)

1. Document Upload
2. Choose Path
    a. Path 1: Self Service (skips Recipient Add)
    b. Path 2: Recipient Based (follows current flow Steps 2–6)
3. Form Creation (Path 1)
4. Bind Form with Self-Service Envelope
5. Embed Form (on Website or Share Form Link)

---

## Current Signing Flow

1. Follow the link received via email.
2. Click on the Sign Field and draw or upload the signature.

---

## Upgraded Signing Flow

Step 1: Access the document link via Email, Embedded Website Form, or other source.

Step 2: Pre-Sign Security & Signature Processing

| Step | Action | Description / Notes |
|---|---|---|
| 2.1 | OTP Challenge | Email / SMS / Phone verification |
| 2.2 | Prepare Signature Placeholder | System adds placeholder in PDF and computes exact byte-range hash |
| 2.3 | Generate Cryptographic Signature | X.509 certificate + CMS format |

| | | |
|---|---|---|
| 2.4 | Timestamping (Optional) | Trusted timestamp token (TSA) added for non-repudiation |
| 2.5 | Save Signed PDF & Progress Envelope | Envelope moves to next signer in order |
| 2.6 | Evidence / Audit Storage | Store docHash, certSerial, actor, timestamp, OTP result, IP/Device info, optional anchorTx / Merkle proof |
| 2.7 | Validation Endpoint | Reconstructs and verifies all stored evidence for compliance |

---

## Layered Flow of the Enhanced Digital Signature System

| Step | What happens | Libraries involved | Why / Purpose |
|---|---|---|---|
| **Upload Document / Create Envelope** | User uploads PDF; empty envelope is created in DB | None at this step | Existing app logic; sets up envelope and tracks metadata |
| **Add Recipients** | Define recipients, roles, order of signing | None directly | Existing logic to handle users and permissions |
| **Place Signature Fields** | Map fields (signature, initials, date, etc.) on the PDF | **pdf-lib** | Prepares the PDF for signing; adds visible signature fields for each recipient |
| **Send Envelope** | Email notification sent to recipients | None directly | Existing mail service triggers; no crypto needed yet |
| **Recipients Sign** | Recipients click in-field to sign PDF | **node-signpdf**, **jsrsasign**, **speakeasy / otplib** | Node-signpdf applies digital signature to PDF; jsrsasign adds timestamp (RFC 3161); speakeasy/otplib handles OTP/TAN for strong authentication |
| **Envelope Completed** | All recipients sign → envelope status updated | **winston / pino** | Write immutable audit trail logs; ensures compliance |

## Crypto Layer (Security & Signing)

| Step | What happens | Libraries | Why |
|---|---|---|---|
| **Key & Certificate Generation** | Generate RSA/ECDSA keys, X.509 certs | **node-forge**, **PKI.js** | Node-forge: create dev/test certs and serial numbers; PKI.js: parsing & validation of certs for compliance |
| **PDF Signing** | Apply signature to PDF | **node-signpdf** | Signs the document in a compliant manner (SES/AdES) |
| **Timestamping** | Record signing time for long-term validation | **jsrsasign (TSP)**, optionally **OpenTimestamps** | Ensures the signature is tamper-evident and legally valid |
| **OTP/TAN Verification** | Ensure signer is authenticated | **speakeasy / otplib** | Provides second-factor authentication; satisfies AES compliance in EU/UK and IT Act in India |

## Infra Layer (Audit, Hashing, Anchoring)

| Step | What happens | Libraries | Why |
|---|---|---|---|
| **Hash Document** | Generate SHA-256/512 of signed document | **hasha**, **Node crypto** | Creates immutable hash for auditing and anchoring |
| **Merkle Root Creation** | Batch multiple document hashes | **merkletreejs** | Efficient way to anchor multiple documents at once |
| **Blockchain Anchoring** | Store Merkle root or hash on Ethereum / Bitcoin | **ethers.js / web3.js**, **bitcoinjs-lib** | Provides immutable, tamper-proof proof of signing for legal compliance |
| **Audit Trail Logging** | Log all events and signing metadata | **winston / pino** | Structured, append-only logs to comply with US ESIGN, EU eIDAS, India IT Act, etc. |

## How it differs from your current flow

| Current Flow | Enhanced Flow with Libraries | Benefits |
|---|---|---|
| Users upload PDF → add recipients → recipients sign → envelope completed | Same basic flow, but **PDF signed digitally, timestamps added, hashes anchored, OTP verification**, audit logs written | Legal compliance (SES, AdES, AES), tamper-proof signatures, stronger authentication, future LTV support |